

Creating Equipment Classes

Project 2 (Milestone 1) – Documentation

ECE 2774 – Computer Analysis of Power Systems

Prepared by

Souvik Das

Trinh Truong Cong

(Group 7)

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1. Introduction and Scope of Milestone 1

The objective of Milestone 1 is to establish the foundational object oriented representation of a power system network. At this stage, the focus is not on power flow calculations or network matrix assembly, but rather on defining clean, well structured equipment classes that accurately represent physical power system components.

The implemented classes serve as data containers. Each class stores the minimum required parameters and attributes that will later be used to construct network matrices such as the bus admittance matrix (Y_{bus}) and to perform system simulations in subsequent milestones.

The following five equipment classes are implemented:

- Bus
- Transformer
- Transmission Line
- Load
- Generator

This milestone emphasizes clarity and explainability of code rather than numerical computation.

2. Overall Design Philosophy

2.1 Data Container Approach

All classes in this milestone are deliberately implemented as data containers, with no impedance, admittance, or network level calculations included. This approach enforces a strict separation between component definition and system analysis, ensuring that each class is responsible only for storing physical and electrical parameters. Such decoupling preserves architectural clarity and modularity, allowing subsequent milestones to introduce network stamping, matrix assembly, and simulation logic incrementally without requiring any refactoring of the foundational class structures.

2.2 Connectivity via Bus Names

As shown in the class diagram, all equipment classes store connectivity information using bus names (`bus1_name`, `bus2_name`) rather than Bus objects. This design choice:

- keeps the classes loosely coupled
- allows flexible mapping between bus names and bus indices when building Ybus later.

2.3 Unique Bus Indexing Strategy

The Bus class assigns a unique integer index to each bus using a class level counter. This index will later serve as the row/column identifier when assembling network matrices.

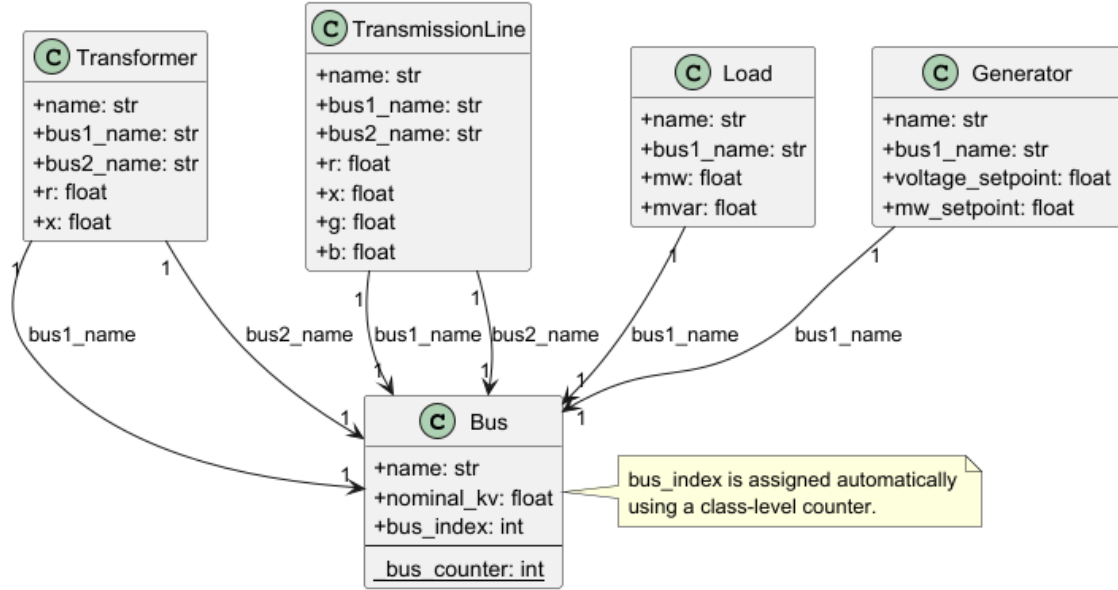


Figure 1: Overview of the Class Diagram

3. Class Diagram Overview

The class diagram illustrates the logical relationships between the equipment classes and the Bus class:

- Transformers and transmission lines connect two buses
- Loads and generators connect to a single bus
- The Bus class acts as the central node to which all equipment is referenced
- Bus indices are assigned automatically using a class level counter

4. Class - by - Class Documentation

This section provides a detailed description of each equipment class implemented in Milestone 1. For every class, the purpose, role of each attribute, and key design decisions

are discussed to ensure complete clarity and explainability. All classes are intentionally designed as data containers that store component parameters and connectivity information, forming the structural foundation for network modeling and analysis in subsequent milestones.

4.1 Bus Class

4.1.1 Purpose

The Bus class represents a node in the power system where electrical components connect. Every piece of equipment in the network ultimately references one or more buses.

4.1.2 Attributes and Their Roles

Table 1: Bus Class Attributes

Attribute	Description
name	Human readable identifier for the bus
nominal_kv	Nominal voltage level of the bus in kV
bus_index	Unique integer index used for matrix construction

4.1.3 Design and Implementation Reasoning

The `bus_index` is assigned automatically using a class level counter, which guarantees that each bus receives a unique identifier without relying on manual indexing. This approach eliminates the risk of indexing errors and keeps the implementation robust as the network scales. The assigned index is deterministic, as it depends solely on the order in which bus objects are created, ensuring repeatable behavior across runs. Using an integer based index also aligns naturally with matrix based formulations employed later in the numerical analysis. Overall, this design allows the Bus class to integrate seamlessly into numerical network models and supports clean, efficient system level computations.

4.2 Transformer Class

4.2.1 Purpose

The Transformer class represents a transformer connecting two buses. In Milestone 1, the transformer is modeled using only its series impedance.

4.2.2 Attributes and Their Roles

Table 2: Transformer Class Attributes

Attribute	Description
name	Transformer identifier
bus1_name	Name of the first connected bus
bus2_name	Name of the second connected bus
r	Series resistance
x	Series reactance

4.2.3 Design and Implementation Reasoning

The transformer model is intentionally kept minimal by storing only the series resistance r and reactance x , without performing any admittance calculations at this stage. More advanced features such as tap changers, phase shifts, and primitive admittance matrices are deliberately excluded to avoid premature complexity. Connectivity is represented using bus names rather than numerical indices, providing greater flexibility during network construction. This design choice ensures that the transformer class can be extended later with additional modeling features without requiring any changes to its existing interface.

4.3 Transmission Line Class

4.3.1 Purpose

The Transmission Line class represents a transmission line connecting two buses using lumped parameters.

4.3.2 Attributes and Their Roles

Table 3: Line Class Attributes and Their Roles

Attribute	Description
name	Line identifier
bus1_name	Sending-end bus
bus2_name	Receiving-end bus
r	Series resistance
x	Series reactance
g	Shunt conductance
b	Shunt susceptance

4.3.3 Design and Implementation Reasoning

The line model includes both series and shunt parameters to capture the essential electrical characteristics while keeping the implementation simple. At this stage, no π model or distributed parameter calculations are performed, and all parameters are stored exactly as provided without transformation. This deliberate choice preserves modeling flexibility and allows the class to be directly extended to π model stamping and more detailed network formulations in future milestones.

4.4 Load Class

4.4.1 Purpose

The Load class represents a constant real and reactive power demand connected to a bus.

4.4.2 Attributes and Their Roles

Table 4: Load Class Attributes

Attribute	Description
name	Load identifier
bus1_name	Bus to which the load is connected
mw	Real power demand (MW)
mvar	Reactive power demand (MVAR)

4.4.3 Design and Implementation Reasoning

Loads are modeled as constant PQ elements, with real and reactive power values stored directly as specified. At this stage, no conversion to admittance or current injection is performed, keeping the representation purely data-driven. This clear separation of data storage from numerical formulation ensures that the load modeling logic remains flexible and can be adapted easily for different analysis types in later stages.

4.5 Generator Class

4.5.1 Purpose

The Generator class represents a generator connected to a bus with specified operating setpoints.

4.5.2 Attributes and Their Roles

Table 5: Generator Class Attributes

Attribute	Description
name	Generator identifier
bus1_name	Connected bus
voltage_setpoint	Voltage magnitude setpoint
mw_setpoint	Real power generation setpoint

4.5.3 Design and Implementation Reasoning

The generator is defined using voltage magnitude and real power setpoints, reflecting a standard PV bus representation. Reactive power output is not computed at this milestone, as the focus is on capturing the generator’s control objectives rather than its full electrical behavior. This formulation aligns directly with conventional PV bus modeling used in later power flow analysis stages.

5. Validation Strategy

A separate validation script is used to verify the correctness of all implemented classes. The script instantiates each class using representative example parameters, prints all required attributes, and confirms correct bus indexing behavior. This validation ensures that constructors function as intended, attributes are stored correctly, and no unintended side effects occur during object creation.

6. Summary

Milestone 1 successfully establishes a clear, extensible object oriented framework for representing power system components. Each class has a well defined purpose, carefully chosen attributes, and a design that supports future analytical development.